Reba.

Vision screening for community health workers in Rwanda.

Five minutes. One cheap phone. No network.

David Mugisho · Ejo Labs Summer Talent Program 2026

THE PROBLEM

Two treatable causes of childhood blindness

- **Retinoblastoma.** A cancer of the eye. Found early, the child lives. Missed, the child dies.
- **Amblyopia.** An eye that never learns to see. By age seven the loss is permanent.

Neither one hurts. The child never complains.

The person who sees the child first is a community health worker, and they have nothing to look with.

ONE SCREENING

Five minutes, start to finish

1. **Consent**, read aloud in the family's language
2. **Vision test**, calibrated against an ID card so the letters are a true size
3. **One photograph of each eye**, flash on
4. **A result**: refer, recheck, or no signs today
5. **A referral slip** to hand to the family

Works offline. Kinyarwanda, English, French, German.

MEASURED, ON THE DEVICE

The red reflex, compared between the two eyes

CLINICAL PHOTOGRAPHS	HEALTHY EYE	AFFECTED EYE
Child 1	0.475	0.381
Child 2	0.476	0.375

Both affected eyes were flagged. Neither healthy eye was. Same phone, same flash, seconds apart, so skin tone and lighting cancel out. **The patient is their own control.**

HONESTY

What is real, and what is not

WORKING TODAY

- Consent and records
- Calibrated visual acuity
- Eye photographs, kept on the phone
- Triage and referral

NOT YET

- A trained model
- Thresholds validated on real eyes
- Kinyarwanda checked by a native speaker

> A model needs thousands of eye photographs labelled by ophthalmologists and taken on phones in villages. That dataset does not exist for Rwanda, because nobody had the tool to collect it. **Reba is that tool.**

WHAT I AM ASKING FOR

Three things

- **A clinical partner** to label the photographs the app is already collecting
- **A pilot** with community health workers in one district
- **A Kinyarwanda speaker** to review the consent script before any real patient